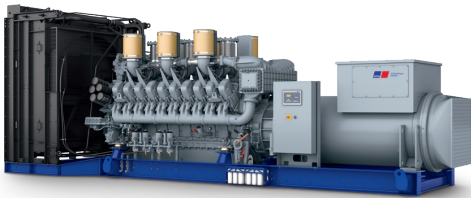




Diesel Generator Set

mtu 20V4000 DS3100

380V – 11 kV/50 Hz/standby power/fuel consumption optimized
20V4000G74F/water charge air cooling



Optional equipment and finishing shown. Standard may vary.

Product highlights

Benefits

- Low fuel consumption
- Optimized system integration ability
- High reliability
- High availability of power
- Long maintenance intervals

Support

- Global product support offered

Standards

- Engine-generator set is designed and manufactured in facilities certified to standards ISO 2008:9001 and ISO 2004:14001
- Generator set complies to ISO 8528
- Generator meets NEMA MG1, BS 5000, ISO, DIN EN and IEC standards
- NFPA 110

Power rating

- System ratings: 3100 kVA - 3200 kVA
- Accepts rated load in one step per NFPA 110*
- Generator set complies to G3 according to ISO 8528-5
- Generator set exceeds load steps according to ISO 8528-5*

Performance assurance certification (PAC)

- Engine-generator set tested to ISO 8528-5 for transient response
- 85% load factor
- Verified product design, quality and performance integrity
- All engine systems are prototype and factory tested

Complete range of accessories available

- Control panel
- Power panel
- Circuit breaker/power distribution
- Fuel system
- Fuel connections with shut-off valve mounted to base frame
- Starting/charging system
- Exhaust system
- Mechanical and electrical driven radiators
- Medium and oversized voltage alternators

Emissions

- Fuel consumption optimized

Certifications

- CE certification option
- Unit certificate acc. to VDE-AR-N 4110

* Changes to the standard parameter sets (alternator-regulator and genset-controller) are necessary



A Rolls-Royce
solution

Application data ¹⁾

Engine			Liquid capacity (lubrication)	
Manufacturer	mtu		Total oil system capacity: l	390
Model	20V4000G74F		Engine jacket water capacity: l	205
Type	4-cycle		Intercooler coolant capacity: l	50
Arrangement	20V		Combustion air requirements	
Displacement: l	95.4			
Bore: mm	170			
Stroke: mm	210		Combustion air volume: m³/s	2.64
Compression ratio	16.4		Max. air intake restriction: mbar	50
Rated speed: rpm	1500		Cooling/radiator system	
Engine governor	ECU 9			
Max power: kWm	2670			
Air cleaner	dry		Coolant flow rate (HT circuit): m3/hr	80
Fuel system			Coolant flow rate (LT circuit): m3/hr	32.5
			Heat rejection to coolant: kW	980
			Heat radiated to charge air cooling: kW	430
			Heat radiated to ambient: kW	105
			Fan power for electr. radiator (40°C): kW	70
Maximum fuel lift: m	5		Exhaust system	
Total fuel flow: l/min	27			
Exhaust gas temp. (after turbocharger): °C				
Fuel consumption ²⁾	l/hr	g/kwh	Exhaust gas volume: m³/s	8.0
At 100% of power rating:	617.6	192	Maximum allowable back pressure: mbar	85
At 75% of power rating:	463.2	192	Minimum allowable back pressure: mbar	30
At 50% of power rating:	321.7	200		

Standard and optional features

System ratings (kW/kVA)

Generator model	Voltage	fuel consumption optimized					
		without radiator			with mechanical radiator		
		kWel	kVA*	AMPS	kWel	kVA*	AMPS
Leroy Somer LSA53.2 M9 (Low voltage Leroy Somer standard)	380 V	2520	3150	4786	2480	3100	4710
	400 V	2520	3150	4547	2480	3100	4474
	415 V	2520	3150	4382	2480	3100	4313
Marathon 1030FDL7094 (Low voltage Marathon)	380 V	2560	3200	4862	2488	3110	4725
	400 V	2552	3190	4604	2488	3110	4489
	415 V	2560	3200	4452	2488	3110	4327
Marathon 1030FDH7101 (Medium volt. marathon)	11kV	2552	3190	167	2488	3110	163
Leroy Somer LSA53.2 ZL14 (Medium volt. Leroy Somer)	11 kV	2560	3200	168	2496	3120	164

* cos phi = 0.8

1 All data refers only to the engine and is based on ISO standard conditions (25°C and 100m above sea level).
2 Values referenced are in accordance with ISO 3046-1. Conversion calculated with fuel density of 0.83 g/ml. All fuel consumption values refer to rated engine power.